PRJ1 Research Gap: Sensor Trust for Lunar South Pole Landing

The maps that predict where descent sensors will mislead a lander already exist, and DEM-based altimeter models have been proven on flight data. We found no lander navigation filter that uses these maps *before descent* to decide how much to trust each sensor. Existing work improves the sensors (SPLICE, NDL, redundant altimeters) or the maps (Barker, Bertone), and treats measurement rejection as a fixed, map-blind rule.

No published method uses the map and lighting to set sensor trust

Columns: how the terrain map is used. Rows: how lighting is handled.

	No map in filter	Map predicts reading	Map sets trust
Lighting sets trust	*none found*	*none found*	**Our trust layer** cliffs, shadows, faults
Planning or hardware	autoNGC (Shoemaker 2022) ShadowNav (Cauligi 2023)	*none found*	*none found*
Lighting ignored	Fixed gate (Bar-Shalom) Adaptive (Mehra 1972) Tomasicchio 2024 Prakash and Raha 2026	SITAN (1983) **Adams 2025 (PITFAL)** Karlgaard 2025 Candan 2026 (craters)	*none found*

where existing methods sit · map use vs lighting

Published work sits in the bottom-left: maps predict readings at best, and lighting is ignored or avoided by scheduling. The top-right cell is the gap this project fills.

The gaps

#	Gap	Evidence	What is missing
G1	Outlier gates are static and terrain-blind	HAKUTO-R M1 rejected a true ~3 km altitude jump at a crater rim; even NASA SPLICE's 2025 PITFAL uses a fixed 99% chi-square limit and inflates noise only after a flag (Adams et al. 2025)	A gate that widens where the DEM predicts a cliff and stays tight elsewhere
G2	Camera trust is not tied to predicted illumination	IM-2's crater matching failed under long shadows; autoNGC avoids bad lighting only by timing images (Shoemaker et al. 2022); crater TRN uses fixed camera noise (Candan and Servadio 2026); polar illumination can be computed in advance (Mazarico et al. 2011, Bertone et al.)	A shadow forecast along the trajectory that lowers camera weight before entering shadow
G3	Sensor faults and real terrain are not told apart	IM-2 altimeter noise, ispace M2 delayed rangefinder and HAKUTO-R valid-but-rejected data all looked like "bad measurements" to the filter	A test that checks a surprising reading against the map before calling it a fault (Prakash D. and Raha 2026 detect inconsistency, but without a map)
G4	Map uncertainty is not used online	PGDA 78 provides per-pixel height and slope error maps; Sheppard et al. 2025 use DEM confidence for hazard maps only, not for filter noise	Using map error to set the expected altimeter noise at each point

#	Gap	Evidence	What is missing
G5	Fixes are reactive and hardware-led	IM-3 and ispace responses add redundant, dissimilar sensors; Chandrayaan-3 relied on a second, wide-footprint altimeter; adaptive filters re-tune only after residuals grow (Mehra 1972)	A software layer that anticipates trouble from the map, usable with existing sensors
G6	Little open, reproducible evaluation	Mission reports are press releases and earnings calls, not datasets; SPLICE and Nova-C filter internals are not public	An open simulation over real South Pole DEMs that others can rerun and watch

G1 and G2 define the core contribution, G3 and G4 refine it, and G5 and G6 make it practical for a student team.

The failures run in both directions. HAKUTO-R (2023) rejected a correct altimeter reading; IM-2 (2025) believed a noisy one, thinking it was ~400 m up when it was about to hit the ground at 10–13 m/s (Aerospace America). A fixed rule cannot get both right; a rule informed by terrain, lighting and consistency over time can.

Site changes are a gap of their own (G7). HAKUTO-R's route was not simulated after its landing site changed. No published method we found re-derives sensor trust automatically when the site or route changes, or checks that it holds at a site the system was never tuned on.

Live scans are not used to judge sensors (G8). Onboard 3D mapping is emerging: SPLICE's hazard-detection lidar maps the landing area during the final descent, and Adams et al. (2025) note that improving lidar now lets landers build terrain maps onboard. In the work we found, these scans serve hazard detection and landing-site navigation. We found no published method that cross-checks the altimeter against both the stored map and a live scan to decide which source to trust. Our optional Extension 1 (live 3D scan) tests this.

Other factors are out of scope. Plume dust (addressed by ARC-LIGHT), boulder avoidance, engine failure and missing sensors also cause failed landings. Our trust layer is built so dust can be added later as another input.

Against the closest paper

	PITFAL (Adams et al., AIAA SciTech 2025)	Our trust layer
DEM in the altimeter prediction	Yes	Yes
Gate size	Fixed, 99% chi-square	Widens with DEM height spread inside the position uncertainty, plus map error
When it acts	After a reading fails	Before descent, along the planned corridor, and on each reading
Fault vs terrain	Consensus across several lidar beams	ML classifier on innovation, terrain and residual features; works with one altimeter
Camera and lighting	Not modelled	Camera weight follows predicted and rendered shadow fraction
Sensors needed	Multi-beam slant-range sensor such as NDL	Single-beam altimeter, camera, IMU
Live 3D scan	Not part of PITFAL; SPLICE's hazard-detection lidar maps the landing area	Optional Extension 1: three-way check of altimeter, stored map and live scan

We say "we found no published method that…" rather than "no method exists": paywalled IEEE Xplore and AIAA ARC proceedings still need a check through the college library.

Key sources

1. ispace (2023). HAKUTO-R Mission 1 debriefing.
2. Adams, D. W., Ward, K. C., Smith, K. W., Pedrotty, S. M., Mendeck, G. F. (2025). Topographic slant range modeling and fault detection for precision planetary landing. AIAA SciTech Forum.
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4. Shoemaker, M. A. et al. (2022). TRN in a lunar landing scenario using autoNGC. AIAA SciTech.

5. Sheppard, A., Reale, C., Skinner, K. A. (2025). A stochastic approach to terrain maps for safe lunar landing. IEEE Aerospace 2026.
6. Prakash D., S. R., Raha, S. (2026). Co-state based data fusion and risk aware filtering. arXiv preprint.
7. Candan, B., Servadio, S. (2026). Deep learning-based lunar crater TRN. arXiv preprint.
8. SpacePolicyOnline (2025). IM-2's imperfect landing and ispace Mission 2.
9. Barker, M. K. et al. (2021). Improved LOLA elevation maps for South Pole landing sites. *Planetary and Space Science* 203, 105119; PGDA 78.
10. Mazarico, E. et al. (2011). Illumination conditions of the lunar polar regions using LOLA topography. *Icarus* 211(2), 1066–1081.
11. Mehra, R. K. (1972). Approaches to adaptive filtering. *IEEE Transactions on Automatic Control* 17(5), 693–698.

Full, annotated list: see the Literature Review document.